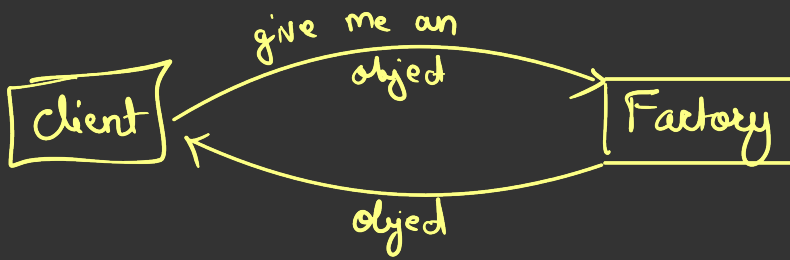


Factory Design Pattern



* Factory Design Pattern

- This pattern decouples object creation code from core Business Logic. → [To avoid complexity]
- The factory design Pattern is a creational design pattern that provides an interface for creating objects while letting subclasses decide which class to instantiate.



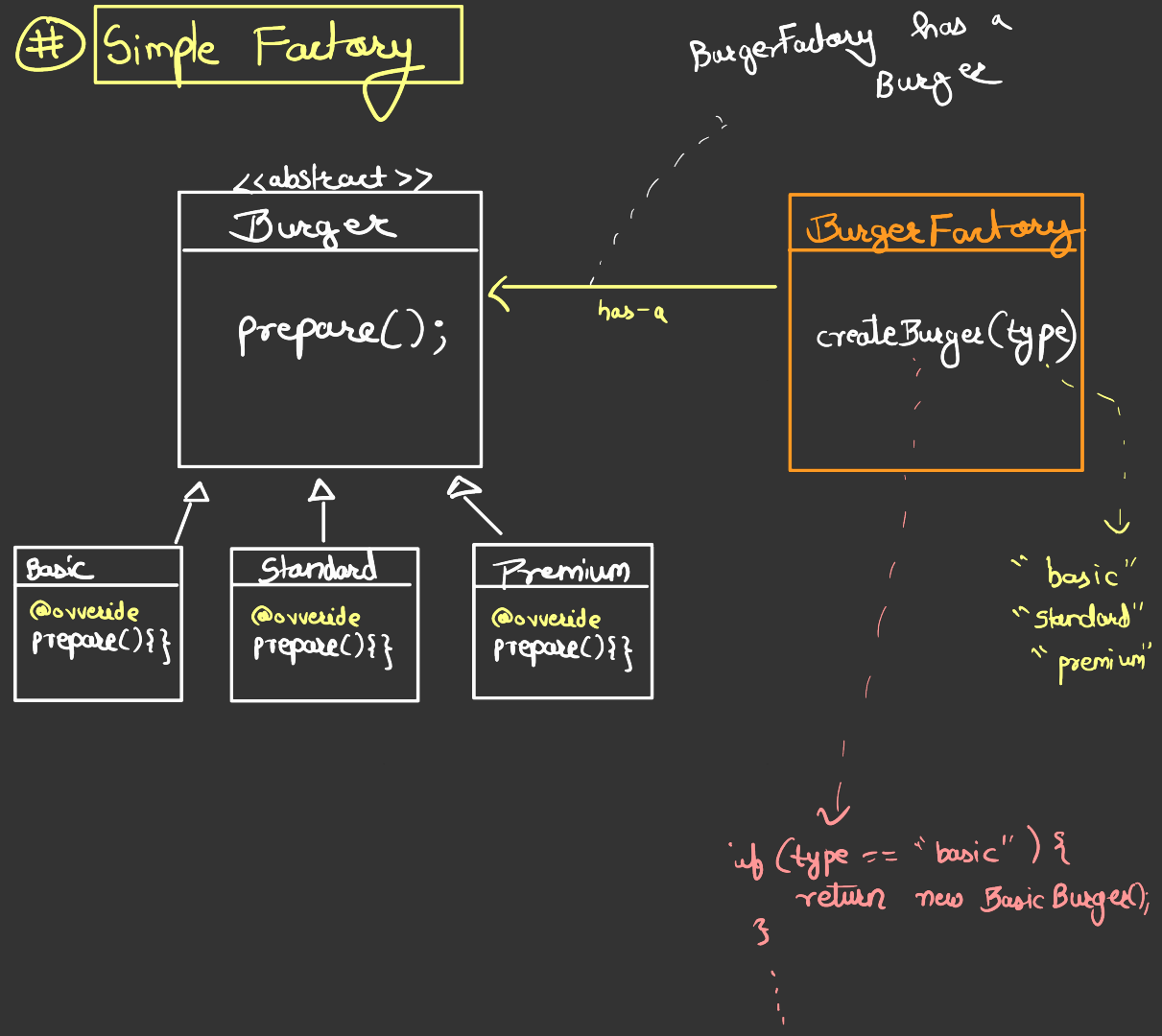
* Factory

→ Humare pass 3 tarike ki factory hoti hai

- ① Simple Factory
 - ② Factory Method
 - ③ Abstract Factory Method
- } extended versions of Simple Factory.

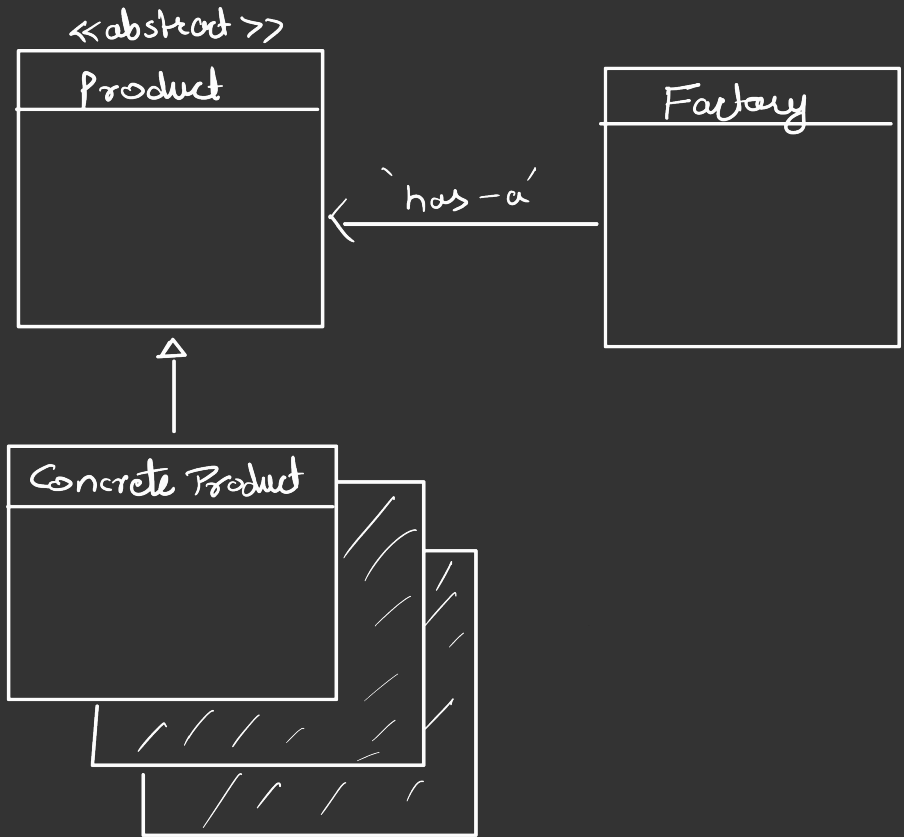
- Simple Factory is not a design Pattern, you can understand it as design Principle but not Pattern.
- **Factory Method** and **Abstract Factory Method** are proper design Pattern.

Simple Factory



this is what simple Factory is!!

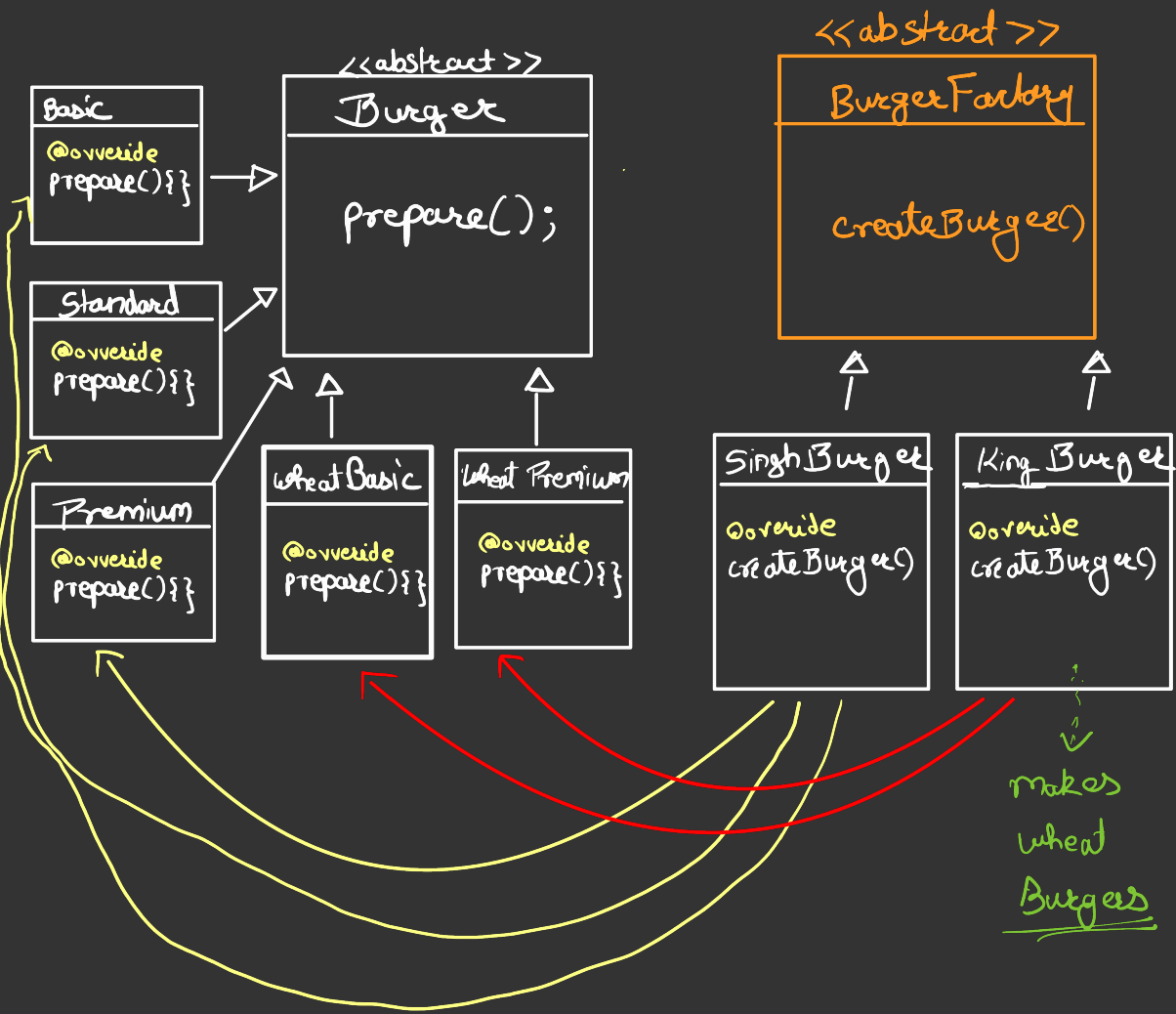
* Standard UML diagram for simple Factory.



• Simple Factory

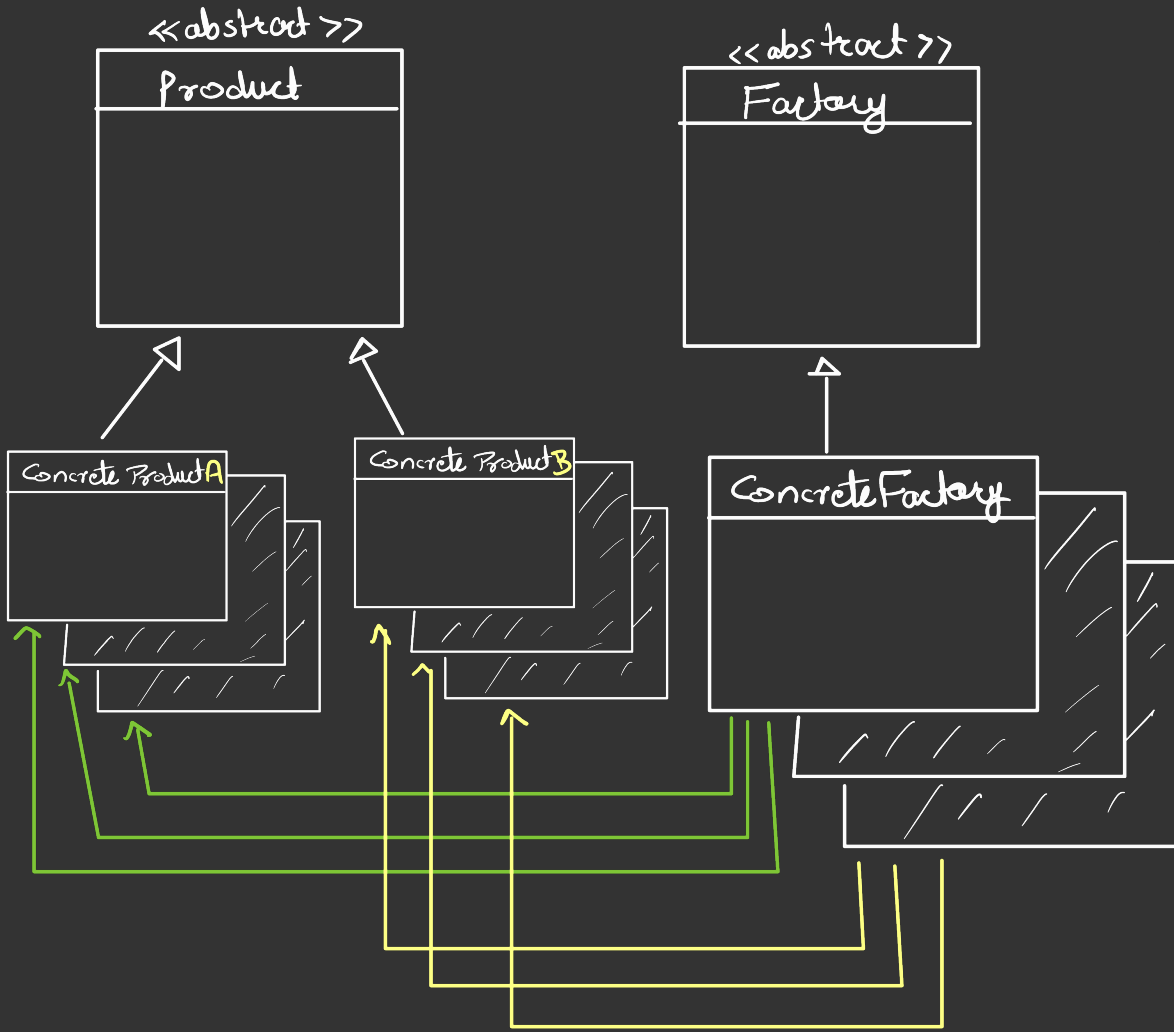
→ A factory class that decides which concrete class to instantiate.

* Factory Method



BurgerFactory factory = new choice of either Singh or King

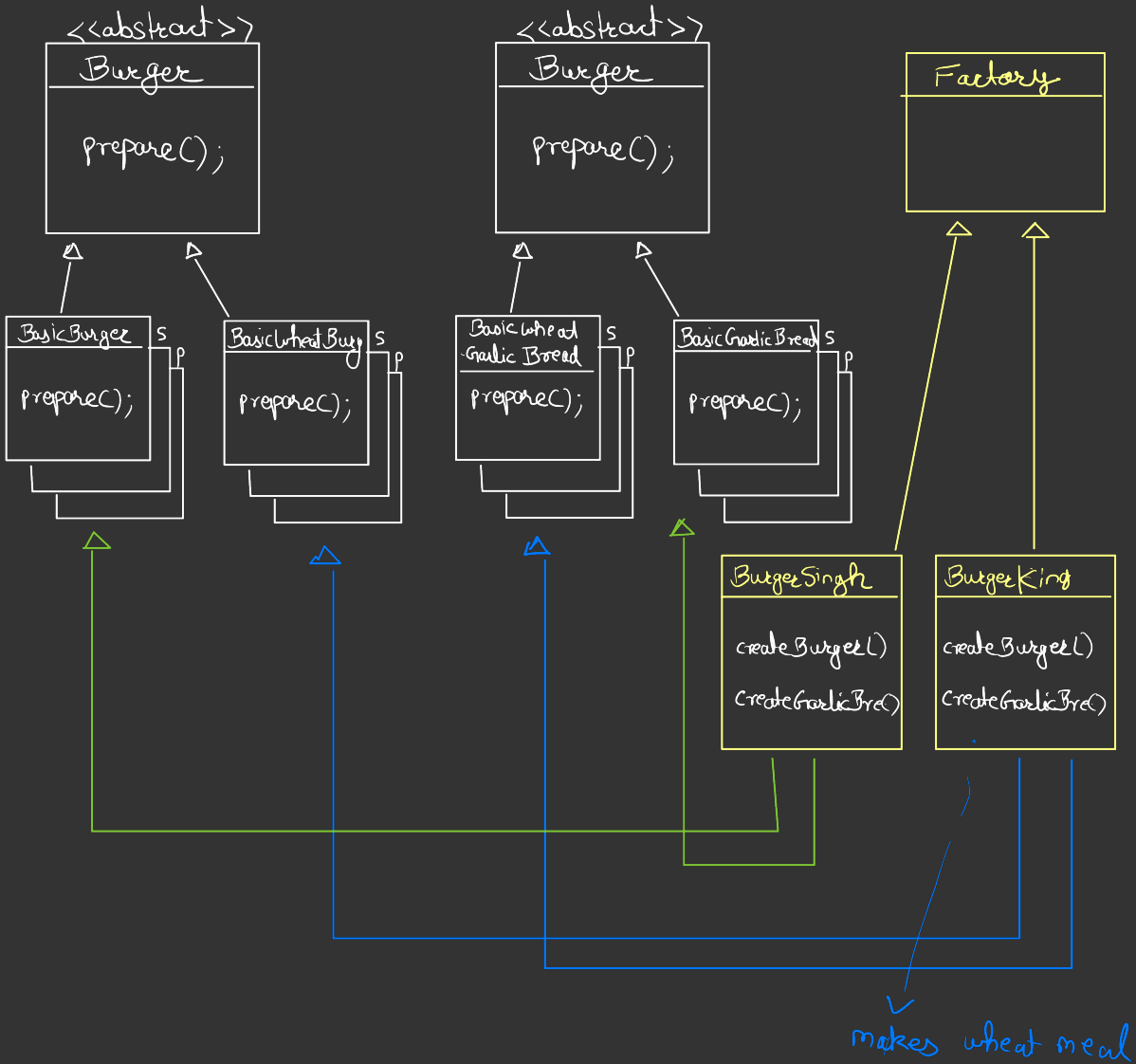
* Standard UML diagram for Factory Method.



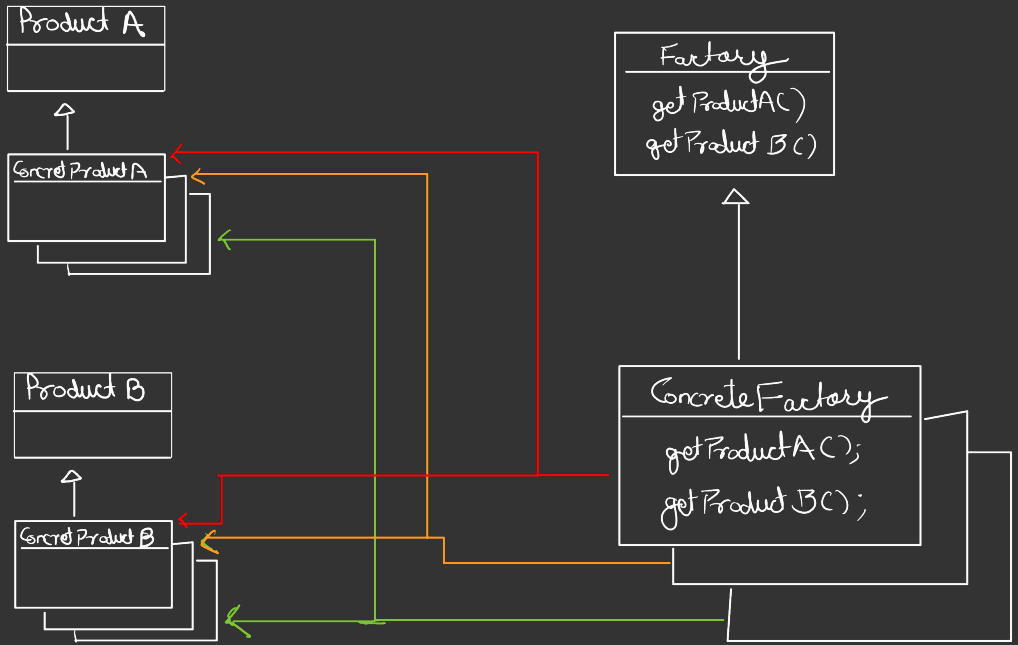
① Factory Method

→ Defines an interface for creating objects but allows subclass to decide which class to instantiate.

* Abstract Factory Method



Standard UML diagram of Abstract Factory Method

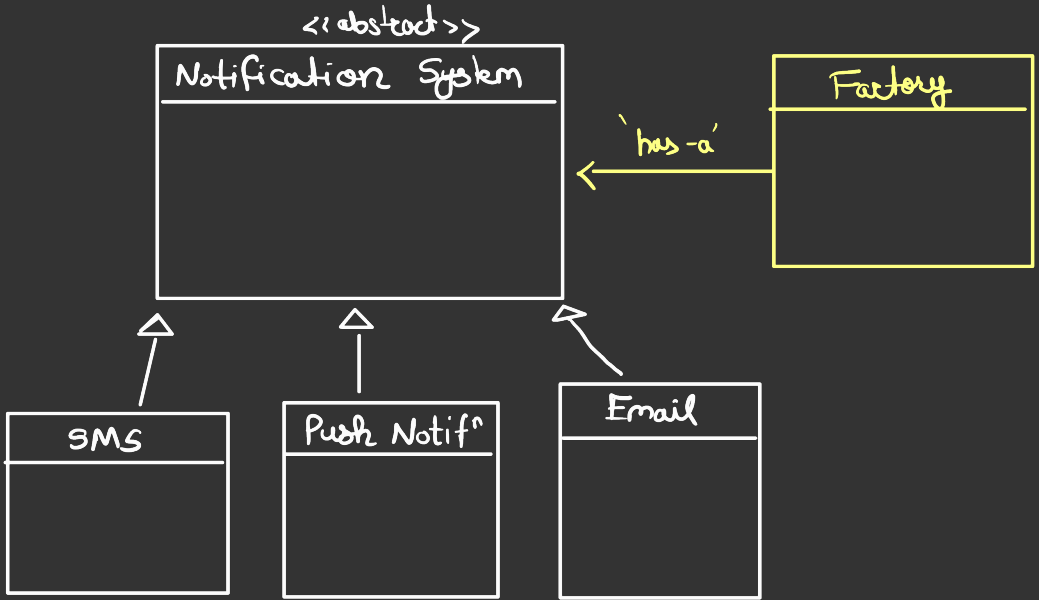


* Abstract Factory Method

→ Provides an interface for creating families of related objects without specifying their concrete classes.

Real World Example

Notification System



↳ Notifⁿ system can be integrated using Strategy design Pattern

